

**ProQuad®****[Measles, Mumps, Rubella and Varicella (Oka/Merck) Virus Vaccine Live]****Refrigerator-stable formulation****I. THERAPEUTIC CLASS**

ProQuad® is a combined attenuated live virus vaccine containing measles, mumps, rubella, and varicella viruses. ProQuad is a sterile lyophilized preparation of (1) the components of M-M-R II\* (Measles, Mumps and Rubella Virus Vaccine Live): Measles Virus Vaccine Live, a more attenuated line of measles virus, derived from Enders' attenuated Edmonston strain and propagated in chick embryo cell culture; Mumps Virus Vaccine Live, the Jeryl Lynn™ (B level) strain of mumps virus propagated in chick embryo cell culture; Rubella Virus Vaccine Live, the Wistar RA 27/3 strain of live attenuated rubella virus propagated in WI-38 human diploid lung fibroblasts; and (2) Varicella Virus Vaccine Live (Oka/Merck), the Oka/Merck strain of varicella-zoster virus propagated in MRC-5 cells (hereafter referred to as VARIVAX\*).

**II. COMPOSITION*****Ila. Active Ingredients***

ProQuad, when reconstituted as directed, is a sterile preparation for intramuscular or subcutaneous administration. Each approximately 0.5-mL dose contains not less than 3.00 log<sub>10</sub> TCID<sub>50</sub> (50% tissue culture infectious dose) of measles virus; 4.30 log<sub>10</sub> TCID<sub>50</sub> of mumps virus; 3.00 log<sub>10</sub> TCID<sub>50</sub> of rubella virus; and a minimum of 3.99 log<sub>10</sub> PFU (plaque-forming units) of Oka/Merck varicella virus.

***Ilb. Inactive Ingredients***

Each approximately 0.5-mL dose of the vaccine contains no more than 20 mg of sucrose, 11 mg of hydrolyzed gelatin, 2.5 mg of urea, 2.3 mg of sodium chloride, 16 mg of sorbitol, 0.38 mg of monosodium L-glutamate, 1.4 mg of sodium phosphate, 0.25 mg of recombinant human

albumin, 0.13 mg of sodium bicarbonate, 94 mcg of potassium phosphate, 58 mcg of potassium chloride; residual components of MRC-5 cells including DNA and protein; 5 mcg of neomycin, bovine serum albumin (0.5 mcg), and other buffer and media ingredients. The product contains no preservative. This product contains trace quantities of ingredients of porcine source.

The cells, virus pools, bovine serum, and recombinant human albumin used in manufacturing are all screened to ensure the absence of adventitious agents.

### III. CLINICAL PHARMACOLOGY

Pharmacotherapeutic group: Viral Vaccine

Vaccine ATC code: J07BD54

Measles, mumps, rubella, and varicella are 4 common childhood diseases caused by measles virus, mumps virus, rubella virus, and varicella virus, respectively. These diseases may be associated with serious complications and/or death. For example, measles can be associated with pneumonia and encephalitis; mumps can be associated with aseptic meningitis, deafness, and orchitis; rubella occurring during pregnancy can cause congenital rubella syndrome in the infants of infected mothers; and wild-type varicella can be associated with bacterial superinfection, pneumonia, encephalitis, and Reye syndrome.

#### *Efficacy*

Formal studies to evaluate the efficacy of ProQuad have not been performed. However, the efficacy of M-M-R II and VARIVAX has been demonstrated in numerous studies.

Efficacy of the measles, mumps, and rubella components of ProQuad was previously established in a series of double-blind controlled field trials with the monovalent vaccines produced by Merck Sharp & Dohme LLC, which demonstrated a high degree of protective efficacy. In these studies seroconversion in response to vaccination against measles, mumps, and rubella paralleled protection from these diseases. ProQuad elicits rates of antibody responses against measles, mumps, and rubella similar to those observed after vaccination with M-M-R II.

More than 518 million doses of M-M-R II have been distributed worldwide (1978 to 2007).

Widespread use of a 2-dose vaccination schedule in the United States and countries such as Finland and Sweden has led to a >99% reduction in the incidence of each of the 3 targeted diseases. Vaccination against measles, mumps, and rubella has led to a significant reduction in the incidence of these diseases.

In combined clinical trials of VARIVAX, the protective efficacy of the vaccine against all forms of varicella ranged from 81 to 100%. In a large case-control study, the vaccine was estimated to be 85% effective against all forms of varicella and 97% effective against moderately severe and severe disease. Long-term estimated efficacy for the vaccine against all forms of varicella over 10 years was 94%. Antibody responses against varicella virus  $\geq 5$  units/mL in the glycoprotein enzyme-linked immunosorbent assay (gpELISA, a highly sensitive assay which is not commercially available) have been shown to be highly correlated with long-term protection. Clinical studies have shown that vaccination with ProQuad elicits rates of antibody responses against varicella virus  $\geq 5$  units/mL in the gpELISA similar to those observed after vaccination with VARIVAX.

#### *Immunogenicity*

Immunogenicity was studied in children 12 through 23 months of age with a negative clinical history of measles, mumps, rubella, and varicella who participated in 5 randomized clinical trials. The immunogenicity of the current refrigerator-stable formulation was shown to be similar to the immunogenicity of the earlier formulation of ProQuad. Clinical trials also established that the earlier formulation of ProQuad is similar to the individual component vaccines (M-M-R II and VARIVAX), which are currently used in routine vaccination in some countries.

Clinical trials involving 6987 subjects who received ProQuad demonstrated detectable immune responses to measles, mumps, rubella, and varicella in a high proportion of individuals. The presence of detectable antibody was assessed by an appropriately sensitive enzyme-linked immunosorbent assay (ELISA) for measles, mumps (wild-type and vaccine-type strains), and rubella, and by gpELISA for varicella. Following a single dose of ProQuad, the vaccine response rates were 97.7% for measles, 96.3 to 98.8% for mumps, and 98.8% for rubella. The vaccine response rate was 90.9% for varicella based on an antibody response rate  $\geq 5$  gpELISA units/mL (a response rate that has been shown to be highly correlated with long-term protection). These results were similar to the immune response rates induced by concomitant administration of M-M-R II and VARIVAX at separate injection sites.

### *Children who received a second dose of ProQuad*

In 2 clinical trials, 1035 subjects were administered a second dose of ProQuad subcutaneously approximately 3 months after the first dose. The vaccine response rates were 99.4% for measles, 99.9% for mumps, 98.3% for rubella, and 99.4% for varicella ( $\geq 5$  gpELISA units/mL). The geometric mean titers (GMTs) following the second dose of ProQuad increased approximately 2-fold each for measles, mumps, and rubella, and approximately 41-fold for varicella. In these trials, the rates of adverse experiences after the second dose of ProQuad were generally similar to, or lower than, those seen with the first dose. The fever rate was lower after the second dose than after the first dose.

### *Children who received ProQuad at 4 through 6 years of age after primary vaccination with M-M-R II and VARIVAX*

The immunogenicity and safety of ProQuad were evaluated in a clinical trial involving 799 subjects 4 through 6 years of age who had received M-M-R II and VARIVAX at least 1 month prior to study entry. Following the dose of ProQuad administered subcutaneously, GMTs for measles, mumps, rubella, and varicella were similar to those following a second dose of M-M-R II and VARIVAX administered concomitantly at separate injection sites. Additionally, GMTs for measles, mumps, and rubella were similar to those following a second dose of M-M-R II given concomitantly with placebo. In this trial, the rates and types of adverse experiences seen in the group that received ProQuad were generally similar to those seen in the control groups.

### *Immunogenicity Following 2 Doses of ProQuad Administered Intramuscularly or Subcutaneously*

In an open label clinical trial (NCT00402831) 405 children 12 through 18 months of age received two doses of ProQuad, administered 30 days apart, either intramuscularly (n=202) or subcutaneously (n=203). Antibody responses to measles, mumps, rubella and varicella viruses were measured by ELISAs using sera obtained 30 days post-dose 1, and 6-weeks post-dose 2. For anti-measles virus, anti-mumps virus, anti-rubella virus and anti-varicella virus, seroresponse rates were defined as the percentage of children seronegative at baseline who achieved antibody titers above the respective seroresponse threshold for each assay 30 days post-dose 1 and 6 weeks post-dose 2. Seroresponse thresholds were defined as 255 mIU/mL, 10 EU/mL, 10 IU/mL and 5 gpELISA units for anti-measles virus, anti-mumps virus, anti-rubella virus and anti-varicella virus antibodies, respectively after each dose. For each vaccine antigen at least 87% of enrolled children were seronegative at baseline.

In the pre-specified primary analysis, the seroresponse rates to measles, mumps, rubella, and varicella viruses after dose 2 were noninferior in the intramuscular group compared to the subcutaneous group (lower bound of the 95% confidence interval for the difference in seroresponse rates [intramuscular group minus subcutaneous group]  $>-10\%$ ). The proportions of children achieving antibody titers above the seroresponse thresholds for measles, mumps, rubella, and varicella viruses after dose 1 were as follows: 100%, 97.4%, 98.4%, and 98.6%, respectively, in the intramuscular group and 97.3%, 91.3%, 100%, and 98.5%, respectively, in the subcutaneous group. The proportions of children achieving antibody titers above the seroresponse thresholds for measles, mumps, rubella and varicella viruses after dose 2 were as follows: 100%, 99.3%, 100%, and 100%, respectively, in the intramuscular group and 100%, 99.3%, 99.2%, and 99.3%, respectively, in the subcutaneous group.

#### *Persistence of Immune Response*

The persistence of antibody at 1 year after vaccination was evaluated in a subset of 2108 subjects who were involved in 1 clinical trial. The antibody persistence rates 1 year post-vaccination in recipients of a single dose of ProQuad were 98.9% (1722/1741) for measles, 96.7% (1676/1733) for mumps, 99.6% (1796/1804) for rubella, and 97.5% (1512/1550) for varicella ( $\geq 5$  gpELISA units/mL).

Experience with M-M-R II demonstrates that antibodies to measles, mumps, and rubella viruses are still detectable in most individuals 11 to 13 years after primary vaccination. In clinical studies involving healthy subjects who received 1 dose of VARIVAX, detectable varicella antibodies were present in most individuals tested for up to 10 years post-vaccination.

#### *Herpes Zoster*

In a clinical trial, 2 cases of herpes zoster were reported in 2108 healthy subjects 12 through 23 months of age who were vaccinated with ProQuad and followed for 1 year. Both cases were unremarkable and no sequelae were reported.

The reported rate of zoster in recipients of VARIVAX appears not to exceed that previously determined in a population-based study of healthy children who had experienced wild-type varicella. In clinical trials, 12 cases of herpes zoster were reported in 9543 vaccinated individuals 12 months through 12 years of age during 84,414 person-years of follow-up. This resulted in a calculated incidence of at least 0.14 cases per 1,000 person-years. The incidence of herpes zoster following naturally acquired infection in subjects  $>5$  years of age and persons 5

to 9 years of age has been reported to be 1.1 and 0.51 per 1,000 person-years, respectively. All 12 cases reported after VARIVAX were mild and no sequelae were reported. The long-term effect of VARIVAX on the incidence of herpes zoster is unknown at present.

#### *Post-Exposure Prophylaxis*

No clinical data are available for ProQuad administered after exposure to measles, mumps, rubella, or varicella; however, post-exposure prophylaxis has been demonstrated for measles and varicella with measles-containing vaccine and varicella virus vaccine, respectively. Vaccination of susceptible individuals within 3 days of exposure to wild-type measles may provide some protection. Vaccination of susceptible individuals within 3 days of exposure to wild-type varicella may prevent a clinically apparent infection or modify the course of the infection. In addition, there are limited data that indicate that vaccination up to 5 days after exposure to varicella may modify the course of the infection.

#### *Reye Syndrome*

Reye syndrome following wild-type varicella infection has occurred in children and adolescents, the majority of whom had received salicylates. In clinical studies of ProQuad and in the clinical studies of VARIVAX, physicians advised subjects not to use salicylates for 6 weeks after vaccination. There were no reports of Reye syndrome in recipients of ProQuad or VARIVAX during these studies.

#### *Studies With Other Vaccines*

In a clinical trial involving 1913 healthy subjects 12 through 15 months of age, 949 received ProQuad subcutaneously, Diphtheria and Tetanus Toxoids and Acellular Pertussis Vaccine Adsorbed (DTaP) and *Haemophilus b* Conjugate (Meningococcal Protein Conjugate) and Hepatitis B (Recombinant) Vaccine concomitantly at separate injection sites. Another 485 healthy subjects received ProQuad at the initial visit followed by DTaP and *Haemophilus b* Conjugate and Hepatitis B (Recombinant) Vaccine given concomitantly 6 weeks later. In subjects 13.5 months of age or older, seroconversion rates and antibody titers were comparable between the 2 groups at approximately 6 weeks post-vaccination. However, in subjects less than 13.5 months of age, seroconversion rates and antibody titers were comparable between the 2 groups for each of the vaccine components except pertussis FHA (see **DRUG INTERACTIONS**). No clinically significant differences in adverse experiences were reported between the 2 treatment groups.

In a clinical trial involving 1027 healthy children 12 to 15 months of age, 510 were randomized to receive ProQuad subcutaneously and Prevnar concomitantly at separate injection sites, and 517 were randomized to receive ProQuad and Prevnar non-concomitantly. Seroconversion rates and antibody titers for measles, mumps, rubella, varicella, and *S. pneumoniae* types 4, 6B, 9V, 14, 18C, 19F, and 23F were comparable in the concomitant and non-concomitant groups at 6 weeks post-vaccination indicating that ProQuad and Prevnar can be administered concomitantly at separate injection sites. No clinically significant differences in adverse events were reported between treatment groups.

In a clinical trial involving 1800 healthy children 12 to 23 months of age, 1453 were randomized to receive 2 doses of VAQTA, and 347 were randomized to receive 2 doses of VAQTA concomitantly with 2 doses ProQuad subcutaneously at least 6 months apart. Rates of adverse experiences were lower following a second dose than following the first dose of both vaccines given concomitantly.

In a clinical trial involving 653 healthy children 12 to 15 months of age, 330 were randomized to receive VAQTA, ProQuad subcutaneously, and Prevnar concomitantly, and 323 were randomized to receive ProQuad subcutaneously and Prevnar concomitantly followed by VAQTA 6 weeks later. Seroconversion rates and antibody titers for measles, mumps, rubella, varicella, and *S. pneumoniae* types 4, 6B, 9V, 14, 18C, 19F, and 23F were comparable between the 3 groups at 6 weeks post-vaccination indicating that ProQuad, VAQTA, and Prevnar can be administered concomitantly at separate injection sites. No clinically significant differences in adverse events were reported among treatment groups.

In the above 3 post-licensure clinical trials evaluating the concomitant use of ProQuad with other pediatric vaccines, a total of 1745 children 12 to 23 months of age received 2 doses of ProQuad, of which 1661 completed safety follow-up after both doses. Rates of adverse experiences after the second dose of ProQuad were generally similar to, or lower than, those seen with the first dose. The fever rate was lower after the second dose than after the first dose.

#### IV. INDICATIONS

ProQuad is indicated for vaccination against measles, mumps, rubella, and varicella in individuals 12 months through 12 years of age.

## V. DOSAGE AND ADMINISTRATION

### *Dosage*

Individuals 12 months through 12 years of age should receive a single dose of ProQuad administered intramuscularly or subcutaneously.

If a second dose of measles-containing vaccine is to be administered according to applicable official recommendations, then ProQuad may be used for this dose.

If the first dose of a measles-containing vaccine is given between 6 months of age and less than 12 months of age (in an at-risk situation such as measles outbreak, or due to official recommendations) the response to the vaccine may be adversely influenced by circulating maternal antibodies. Therefore, another dose of a measles-containing vaccine should be given at 12 months of age or later. A subsequent (third) dose can be administered if warranted by official recommendations for a measles-containing vaccine.

At least 1 month should elapse between a dose of M-M-R II and ProQuad. If a second dose of varicella-containing vaccine is administered, there should be a minimum interval of 3 months between doses.

Do not give immune globulin (IG) or Varicella-Zoster Immune Globulin (VZIG) concomitantly with ProQuad (see **DRUG INTERACTIONS**).

### *Method of Administration*

FOR INTRAMUSCULAR OR SUBCUTANEOUS ADMINISTRATION ONLY. DO NOT INJECT INTRAVASCULARLY.

The vaccine is to be injected in the deltoid region of the upper arm or in the higher anterolateral area of the thigh.

**CAUTION:** A sterile syringe free of preservatives, antiseptics, detergents, and other antiviral substances must be used for each injection and/or reconstitution of ProQuad because these substances may inactivate the vaccine viruses.

To reconstitute the vaccine, use only the diluent supplied because it is free of preservatives or other antiviral substances, which might inactivate the vaccine viruses.

It is important to use a separate sterile syringe and needle for each patient to prevent transmission of infectious agents from one individual to another.

Withdraw the entire volume of solvent into a syringe (if a prefilled syringe is available, this step is not necessary). Inject the entire content of the syringe into the vial containing the powder. Gently agitate to dissolve completely. Withdraw the entire content of the reconstituted vaccine from the vial into the same syringe and inject the entire volume.

**IT IS RECOMMENDED THAT THE VACCINE BE ADMINISTERED IMMEDIATELY AFTER RECONSTITUTION, TO MINIMIZE LOSS OF POTENCY. DISCARD IF RECONSTITUTED VACCINE IS NOT USED WITHIN 30 MINUTES.**

Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit. Before reconstitution, the lyophilized vaccine is a white to pale yellow compact crystalline plug. ProQuad, when reconstituted, is a pale yellow to light pink liquid.

## **VI. CONTRAINDICATIONS**

History of hypersensitivity to any component of the vaccine, including gelatin.

History of anaphylactoid reaction to neomycin.

Blood dyscrasias, leukemia, lymphomas of any type, or other malignant neoplasms affecting the bone marrow or lymphatic system.

Immunosuppressive therapy (including high-dose corticosteroids); however, ProQuad is not contraindicated for use in individuals who are receiving topical corticosteroids or low-dose corticosteroids, as are commonly used for asthma prophylaxis or in patients who are receiving corticosteroids as replacement therapy, e.g., for Addison's disease. Vaccination with a live attenuated vaccine, such as varicella, can result in a more extensive vaccine-associated rash or disseminated disease in individuals on immunosuppressant doses of corticosteroids. Individuals who are on immunosuppressant drugs are more susceptible to infections than healthy individuals.

Primary and acquired immunodeficiency states, including immunosuppression in association

with AIDS or other clinical manifestations of infection with human immunodeficiency viruses; cellular immune deficiencies; and hypogammaglobulinemic and dysgammaglobulinemic states. Measles inclusion body encephalitis, pneumonitis, and death as a direct consequence of disseminated measles vaccine virus infection have been reported in severely immunocompromised individuals inadvertently vaccinated with measles-containing vaccine.

Family history of congenital or hereditary immunodeficiency, unless the immune competence of the potential vaccine recipient is demonstrated.

Active untreated tuberculosis.

Any active febrile illness with fever  $>38.5^{\circ}\text{C}$  ( $>101.3^{\circ}\text{F}$ ); however, low-grade fever itself is not a contraindication to vaccination.

Do not give ProQuad to pregnant females. If vaccination of postpubertal females is undertaken, pregnancy should be avoided for 3 months following vaccination (see **PREGNANCY**).

## VII. PRECAUTIONS

### *General*

Adequate treatment provisions, including epinephrine injection (1:1000), should be available for immediate use should an anaphylactic or anaphylactoid reaction occur.

Due caution should be employed in administration of ProQuad to persons with individual or family history of convulsions, a history of cerebral injury or any other condition in which stress due to fever should be avoided. The physician should be alert to the temperature elevation that may occur following vaccination (see **SIDE EFFECTS**).

The safety and efficacy of ProQuad have not been established in individuals who are known to be infected with human immunodeficiency viruses with or without evidence of immunosuppression (see **CONTRAINDICATIONS**).

The duration of protection from measles, mumps, rubella, and varicella infection after vaccination with ProQuad is unknown (see **CLINICAL PHARMACOLOGY**).

As for any vaccine, vaccination with ProQuad may not result in protection in all vaccine

recipients.

### *Transmission*

Excretion of small amounts of the live attenuated rubella virus from the nose or throat has occurred in the majority of susceptible individuals 7 to 28 days after vaccination. There is no confirmed evidence to indicate that such virus is transmitted to susceptible persons who are in contact with the vaccinated individuals. Consequently, transmission through close personal contact, while accepted as a theoretical possibility, is not regarded as a significant risk. However, transmission of the rubella vaccine virus to infants via breast milk has been documented (see **NURSING MOTHERS**).

There are no reports of transmission of the more attenuated Enders' Edmonston strain of measles virus or the Jeryl Lynn™ strain of mumps virus from vaccine recipients to susceptible contacts.

Post-licensing experience suggests that transmission of varicella vaccine virus (Oka/Merck) resulting in varicella infection including disseminated disease may occur rarely between vaccine recipients (who develop or do not develop a varicella-like rash) and contacts susceptible to varicella including healthy as well as high-risk individuals.

High-risk individuals susceptible to varicella include:

- Immunocompromised individuals (see **CONTRAINDICATIONS**);
- Pregnant women without documented positive history of varicella (chickenpox) or laboratory evidence of prior infection;
- Newborn infants of mothers without documented positive history of varicella or laboratory evidence of prior infection.

Vaccine recipients should attempt to avoid, whenever possible, close association with high-risk individuals susceptible to varicella for up to 6 weeks following vaccination. In circumstances where contact with high-risk individuals susceptible to varicella is unavoidable, the potential risk of transmission of the varicella vaccine virus should be weighed against the risk of acquiring and transmitting wild-type varicella virus.

### *Hypersensitivity to Eggs*

Live measles vaccine and live mumps vaccine are produced in chick embryo cell culture. Persons with a history of anaphylactic, anaphylactoid, or other immediate reactions (e.g., hives,

swelling of the mouth and throat, difficulty breathing, hypotension, or shock) subsequent to egg ingestion may be at an enhanced risk of immediate-type hypersensitivity reactions after receiving vaccines containing traces of chick embryo antigen. The potential risk-to-benefit ratio should be carefully evaluated before considering vaccination in such cases. Such individuals may be vaccinated with extreme caution, having adequate treatment on hand should a reaction occur.

### *Thrombocytopenia*

No clinical data are available regarding the development or worsening of thrombocytopenia in individuals vaccinated with ProQuad. Cases of thrombocytopenia have been reported in post-marketing experience after primary vaccination with ProQuad. In addition, cases of thrombocytopenia have been reported after primary vaccination or revaccination with measles vaccine; with measles, mumps, and rubella vaccine; and with varicella vaccine. Post-marketing experience with live measles, mumps, and rubella vaccine indicates that individuals with current thrombocytopenia may develop more severe thrombocytopenia following vaccination. In addition, individuals who experienced thrombocytopenia following the first dose of a live measles, mumps, and rubella vaccine may develop thrombocytopenia with repeat doses. Serologic status may be evaluated to determine whether or not additional doses of vaccine are needed. The potential risk-to-benefit ratio should be carefully evaluated before considering vaccination with ProQuad in such cases (see **SIDE EFFECTS**).

### *Post-Exposure Prophylaxis*

No clinical data are available for ProQuad administered after exposure to measles, mumps, rubella, or varicella. However, post-exposure prophylaxis has been demonstrated for measles and varicella with a measles-containing vaccine and varicella-containing vaccine, respectively, when administered to the susceptible individuals within 3 days of exposure.

### *Females of Childbearing Age*

In females of childbearing age, pregnancy should be avoided for 3 months following vaccination (see **PREGNANCY**).

### *Adolescents and Adults*

No clinical data are available on the safety, immunogenicity, and efficacy of ProQuad in adolescents and adults.

### *Tuberculin Test*

It has been reported that live attenuated measles, mumps, and rubella virus vaccines given individually may result in a temporary depression of tuberculin skin sensitivity. Therefore, if a tuberculin test is to be done, it should be administered either any time before, simultaneously with, or at least 4 to 6 weeks after ProQuad.

### *Tuberculosis*

Children under treatment for tuberculosis have not experienced exacerbation of the disease when vaccinated with live measles virus vaccine; no studies have been reported to date of the effect of measles virus vaccines on children with untreated tuberculosis.

## **VIII. PREGNANCY**

Studies have not been conducted with ProQuad in pregnant women. ProQuad should not be administered to pregnant females; furthermore, pregnancy should be avoided for 3 months following vaccination (see **INDICATIONS, CONTRAINDICATIONS** and **PRECAUTIONS**).

In counseling women who are inadvertently vaccinated when pregnant or who become pregnant within 3 months of vaccination, the physician should be aware of the following: (1) Reports have indicated that contracting wild-type measles during pregnancy enhances fetal risk. Increased rates of spontaneous abortion, stillbirth, congenital defects and prematurity have been observed subsequent to wild-type measles during pregnancy. There are no adequate studies of the attenuated (vaccine) strain of measles virus in pregnancy. However, it would be prudent to assume that the vaccine strain of virus is also capable of inducing adverse fetal effects; (2) Mumps infection during the first trimester of pregnancy may increase the rate of spontaneous abortion. Although mumps vaccine virus has been shown to infect the placenta and fetus, there is no evidence that it causes congenital malformations in humans; (3) Wild-type rubella infection during pregnancy, especially in the first trimester, can lead to miscarriage, stillbirth, or Congenital Rubella Syndrome (CRS). In an 18-year survey involving over 1200 pregnant women who received rubella vaccine within 3 months before or after conception (of whom 683 received the Wistar RA 27/3 strain), none of the newborns had abnormalities compatible with CRS. Subsequent post-marketing surveillance has identified CRS associated with a rubella vaccine strain following inadvertent vaccination of a pregnant female with a measles, mumps, and rubella vaccine; and (4) Wild-type varicella can sometimes cause harm to the fetus.

## **IX. NURSING MOTHERS**

It is not known whether measles, mumps, or varicella virus is secreted in human milk. Studies have shown that lactating postpartum women vaccinated with live attenuated rubella vaccine may secrete the virus in breast milk and transmit it to breast-fed infants. In the infants who developed serological evidence of rubella infection, none exhibited severe disease; however, one exhibited mild clinical illness typical of acquired rubella. Therefore, caution should be exercised if ProQuad is inadvertently administered to a nursing woman.

## **X. PEDIATRIC USE**

ProQuad has not been studied in infants less than 12 months of age and is not recommended for administration in this age group.

## **XI. DRUG INTERACTIONS**

At least 1 month should elapse between a dose of M-M-R II and a dose of ProQuad. If for any reason a second dose of varicella-containing vaccine is required, at least 3 months should elapse between administration of the 2 doses.

Administration of immune globulins (IG) concomitantly with ProQuad may interfere with the expected immune response. Vaccination should be deferred for at least 3 months following blood or plasma transfusions, or administration of IG. However, the appropriate suggested interval between transfusion or IG administration and vaccination will vary with the type of transfusion or indication for, and dose of, IG (e.g., 5 months for VZIG).

Following administration of ProQuad, any IG including VZIG should not be given for 1 month thereafter unless its use outweighs the benefits of vaccination.

Vaccine recipients should avoid use of salicylates for 6 weeks after vaccination with ProQuad as Reye syndrome has been reported following the use of salicylates during wild-type varicella infection.

The fourth dose of DTaP (diphtheria, tetanus, acellular pertussis vaccine) is indicated for children 15 months of age and older. Limited data suggest that ProQuad may be administered concomitantly (at separate injection sites) with DTaP in children 15 months of age and older (for children less than 15 months of age see **CLINICAL PHARMACOLOGY**).

Results from clinical studies indicate that ProQuad may be administered concomitantly with *Haemophilus b* conjugate (meningococcal protein conjugate), hepatitis B (recombinant), pneumococcal conjugate, and hepatitis A (inactivated) vaccines.

There are no data for the administration of ProQuad with inactivated poliovirus vaccine.

There are insufficient data to support the use of ProQuad with any other vaccines.

## **XII. SIDE EFFECTS**

### *Children 12 through 23 months of age*

In clinical trials, ProQuad was administered subcutaneously to 6038 children 12 through 23 months of age without concomitant administration. ProQuad was generally well tolerated.

Children received either the refrigerator-stable formulation or the frozen formulation of ProQuad and were monitored for 6 weeks post-vaccination. The safety profiles were similar for the two formulations. The safety of the frozen formulation of ProQuad was compared with the safety of M-M-R II and VARIVAX given concomitantly at separate injection sites. The safety profile for ProQuad was similar to the component vaccines.

The only systemic vaccine-related adverse experiences that were reported at a significantly greater rate in individuals who received ProQuad than in individuals who received M-M-R II and VARIVAX concomitantly at separate injection sites were fever ( $\geq 38.9^{\circ}\text{C}$  [ $\geq 102^{\circ}\text{F}$ ] oral equivalent or abnormal) (21.5% versus 14.9%, respectively), and measles-like rash (3.0% versus 2.1%, respectively). Both fever and measles-like rash usually occurred within 5 to 12 days following the vaccination, were of short duration, and resolved with no long-term sequelae. Pain/tenderness/soreness at the injection site was reported at a statistically lower rate in individuals who received ProQuad than in individuals who received M-M-R II and VARIVAX concomitantly at separate injection sites (22.0% versus 26.7%, respectively). The only vaccine-related injection site adverse experience that was more frequent among recipients of ProQuad

than recipients of M-M-R II and VARIVAX was rash at the injection site (2.3% versus 1.5%, respectively).

Across clinical studies, the following adverse experiences were reported as vaccine-related by the investigator in individuals after a single dose of ProQuad (excluding single events with a frequency  $\leq 0.02\%$ ). Several adverse experiences were solicited in the clinical studies and are designated with the symbol (<sup>†</sup>).

[Very common ( $\geq 1/10$ ); Common ( $\geq 1/100$ ,  $< 1/10$ ); Uncommon ( $\geq 1/1,000$ ,  $< 1/100$ ); Rare ( $\geq 1/10,000$ ,  $< 1/1,000$ )]

#### *Infections and infestations*

Common: upper respiratory infection

Uncommon: gastroenteritis, ear infection/otitis, nasopharyngitis, otitis media, pharyngitis, viral infection, viral rash

Rare: tonsillitis, varicella<sup>†</sup>, viral gastroenteritis

#### *Blood and lymphatic disorders*

Rare: lymphadenopathy

#### *Immune system disorders*

Rare: allergy/hypersensitivity

#### *Metabolism and nutrition disorders*

Uncommon: anorexia, decreased appetite

#### *Psychiatric disorders*

Common: irritability

Uncommon: crying, insomnia, sleep disorder

Rare: agitation, clinging

#### *Nervous system disorders*

Uncommon: febrile seizure, somnolence

Rare: ataxia, headache, lethargy

#### *Eye disorders*

Rare: conjunctivitis, tearing, visual discomfort

#### *Ear and labyrinth disorders*

Rare: ear pain

#### *Vascular disorders*

Rare: flushing

#### *Respiratory, thoracic, and mediastinal disorders*

Uncommon: cough, nasal congestion, respiratory congestion, rhinorrhea

Rare: wheezing

#### *Gastrointestinal disorders*

Common: diarrhea, vomiting

Rare: nausea

#### *Skin and subcutaneous tissue disorders*

Common: measles-like rash<sup>†</sup>, rash, varicella-like rash<sup>†</sup>

Uncommon: dermatitis (including contact and atopic), erythema, rubella-like rash<sup>†</sup>, urticaria, viral exanthema

Rare: drug eruption, exanthema

#### *General disorders and administration site conditions*

Very common: fever  $\geq 38.9^{\circ}\text{C}$  ( $\geq 102^{\circ}\text{F}$ ) oral equivalent or abnormal<sup>†</sup>, erythema<sup>†</sup> or pain/tenderness/soreness<sup>†</sup> at the injection site

Common: ecchymosis or swelling<sup>†</sup> at the injection site, injection site rash<sup>†</sup>

Uncommon: asthenia/fatigue, induration or warmth at the injection site, injection site hemorrhage, injection site mass/lump, malaise

Rare: flu-like/influenza-like illness, injection site discoloration, injection site reaction, pain, pain/tenderness/soreness

#### *Injury and poisoning, and procedural complications*

Rare: contusion

#### *Other Adverse Experiences*

Additionally, adverse experiences reported with post-marketing use of ProQuad and/or in clinical studies and/or post-marketing use of M-M-R II, the component vaccines, and VARIVAX without regard to causality or frequency are summarised below.

### *Infections and infestations*

atypical measles, cellulitis, epididymitis, herpes zoster<sup>‡</sup>, infection, measles, orchitis, parotitis, respiratory infection, skin infection, varicella (vaccine strain)

### *Blood and the lymphatic system disorders*

aplastic anemia, lymphadenitis, regional lymphadenopathy, thrombocytopenia

### *Immune system disorders*

anaphylactoid reaction, anaphylaxis and related phenomenon such as angioneurotic edema, facial edema, and peripheral edema, anaphylaxis in individuals with or without an allergic history

### *Psychiatric disorders*

apathy

### *Nervous system disorders*

acute disseminated encephalomyelitis (ADEM), afebrile convulsions or seizures, aseptic meningitis (see below), Bell's palsy, cerebrovascular accident, dizziness, encephalitis<sup>‡</sup> (see below), encephalopathy (see below), Guillain-Barré syndrome, hypersomnia, measles inclusion body encephalitis (see **CONTRAINDICATIONS**), meningitis<sup>‡</sup>, ocular palsies, paraesthesia, polyneuritis, polyneuropathy, subacute sclerosing panencephalitis (see below), syncope, transverse myelitis, tremor

<sup>‡</sup> Cases caused by wild-type varicella or vaccine strain varicella have been reported in immunocompromised and immunocompetent individuals administered VARIVAX (same varicella vaccine strain as in ProQuad).

### *Eye disorders*

edema of the eyelid, irritation, necrotizing retinitis (reported only in immunocompromised individuals), optic neuritis, retinitis, retrobulbar neuritis

### *Ear and labyrinth disorders*

nerve deafness

### *Vascular disorders*

extravasation

### *Respiratory, thoracic and mediastinal disorders*

bronchial spasm, bronchitis, pneumonitis (see **CONTRAINDICATIONS**), pneumonia, pulmonary congestion, rhinitis, sinusitis, sneezing, sore throat

*Gastrointestinal disorders*

abdominal pain, hematochezia, mouth ulcer

*Skin and subcutaneous tissue disorders*

erythema multiforme, Henoch-Schönlein purpura, panniculitis, pruritus, purpura, skin induration, Stevens-Johnson syndrome, acute hemorrhagic edema of infancy, skin granuloma associated with vaccine derived rubella virus

*Musculoskeletal, connective tissue and bone disorders*

arthritis and/or arthralgia (usually transient and rarely chronic [see below]), musculoskeletal pain, myalgia, swelling

*General disorders and administration site conditions*

injection site complaints (burning and/or stinging of short duration, edema/swelling, hive-like rash, hematoma, induration, lump, vesicles, wheal and flare), inflammation, papillitis, stiffness, varicella-like rash, warm sensation, warm to touch

Death from various, and in some cases unknown, causes has been reported rarely following vaccination with measles, mumps, and rubella vaccines; however, a causal relationship has not been established in healthy individuals (see **CONTRAINDICATIONS**). No deaths or permanent sequelae were reported in a published post-marketing surveillance study in Finland involving 1.5 million children and adults who were vaccinated with M-M-R II during 1982 to 1993.

Encephalitis and encephalopathy have been reported approximately once for every 3 million doses of the combination of measles, mumps, and rubella vaccine contained in M-M-R II. Since 1978, post-marketing surveillance of M-M-R II indicates that serious adverse events such as encephalitis and encephalopathy continue to be rarely reported. The risk of such serious neurological disorders following live measles virus vaccine administration remains far less than that for encephalitis and encephalopathy with wild-type measles (1 per 1000 reported cases).

In severely immunocompromised individuals inadvertently vaccinated with measles-containing vaccine, measles inclusion body encephalitis, pneumonitis, and fatal outcome as a direct consequence of disseminated measles vaccine virus infection have been reported (see

**CONTRAINDICATIONS**); disseminated mumps and rubella vaccine virus infection have also been reported.

Arthralgia and/or arthritis (usually transient and rarely chronic), and polyneuritis are features of infection with wild-type rubella and vary in frequency and severity with age and gender, being greatest in adult females and least in prepubertal children. Following vaccination in children, reactions in joints are generally uncommon (0 to 3%) and of brief duration. In women, incidence rates for arthritis and arthralgia are generally higher than those seen in children (12 to 20%), and the reactions tend to be more marked and of longer duration. Symptoms may persist for a matter of months or on rare occasions for years. In adolescent girls, the reactions appear to be intermediate in incidence between those seen in children and adult women. Even in older women (35 to 45 years), these reactions are generally well tolerated and rarely interfere with normal activities.

Chronic arthritis has been associated with wild-type rubella infection and has been related to persistent virus and/or viral antigen isolated from body tissues. Only rarely have vaccine recipients developed chronic joint symptoms.

There have been reports of subacute sclerosing panencephalitis (SSPE) in children who did not have a history of infection with wild-type measles but did receive measles vaccine. Some of these cases may have resulted from unrecognized measles in the first year of life or possibly from the measles vaccination. Based on estimated measles vaccine distribution in the United States (US), the association of SSPE cases to measles vaccination is about one case per million vaccine doses distributed. This is far less than the association with infection with wild-type measles, 6 to 22 cases of SSPE per million cases of measles. The results of a retrospective case-controlled study conducted by the US Centers for Disease Control and Prevention suggest that the overall effect of measles vaccine has been to protect against SSPE by preventing measles with its inherent higher risk of SSPE.

Cases of aseptic meningitis have been reported following measles, mumps, and rubella vaccination. Although a causal relationship between other strains of mumps vaccine and aseptic meningitis has been shown, there is no evidence to link Jeryl Lynn™ mumps vaccine to aseptic meningitis.

In a randomized open-label clinical trial (NCT00402831), conducted in France, 405 children 12 months through 18 months of age received 2 doses of ProQuad, administered 30 days apart

(a non-US licensed interval), by either the intramuscular (n=202) or subcutaneous (n=203) route. In the overall population, 50.9% were male and the median age was 13.3 months. Local and systemic solicited adverse reactions were recorded by parents or guardians using standardized diary cards. Local solicited reactions were recorded for 4 days after vaccination, and systemic solicited adverse reactions were recorded for 28 days after vaccination. In the event that a participant experienced a rash or a mumps-like illness, parents and/or guardians were instructed to contact the investigator for an examination as soon as possible and no later than 72 hours following onset of symptoms. The nature of any rash was characterized by principal investigator either as measles-like, rubella-like, varicella-like or “other”. Study investigators reviewed the diary card with the participant or participant’s legal guardian 30 days and 42 days after dose 1 and dose 2, respectively to ensure consistency with protocol definitions. Tables 1 and 2 below present the frequency of solicited adverse reactions based on the final assessment by the study investigators.

**Table 1: Proportion of Participants Reporting Solicited Adverse Reactions Following First Vaccination with ProQuad, by the Intramuscular or Subcutaneous Route**

|                                                                    | INTRAMUSCULAR<br>N=202<br>% | SUBCUTANEOUS<br>N=203<br>% |
|--------------------------------------------------------------------|-----------------------------|----------------------------|
| Solicited local reactions at ProQuad injection site (Days 0 to 4)* |                             |                            |
| Erythema†                                                          | 5.0                         | 14.3                       |
| Mild                                                               | 4.5                         | 12.8                       |
| Moderate                                                           | 0                           | 1.5                        |
| Severe                                                             | 0                           | 0                          |
| Missing                                                            | 0.5                         | 0                          |
| Pain‡                                                              | 10.9                        | 5.9                        |
| Mild                                                               | 8.9                         | 3.9                        |
| Moderate                                                           | 2.0                         | 2.0                        |
| Severe                                                             | 0                           | 0                          |
| Swelling†                                                          | 1.0                         | 3.9                        |
| Mild                                                               | 1.0                         | 3.9                        |
| Moderate                                                           | 0                           | 0                          |

|                                                                 |      |      |
|-----------------------------------------------------------------|------|------|
| Severe                                                          | 0    | 0    |
| Solicited systemic reactions (Days 0 to 28)                     |      |      |
| Measles-like rash                                               | 0.5  | 2.0  |
| Rubella-like rash                                               | 3.0  | 3.0  |
| Varicella-like rash                                             | 1.0  | 0.5  |
| Zoster-like rash                                                | 0    | 0    |
| Mumps-like illness                                              | 0.5  | 0    |
| Fever (temperature $\geq 38.0^{\circ}\text{C}$ ) <sup>§,¶</sup> | 62.8 | 68.3 |
| 38.0-38.5°C                                                     | 21.1 | 17.6 |
| >38.5-39.0°C                                                    | 18.1 | 21.6 |
| >39.0-39.5°C                                                    | 14.1 | 18.1 |
| >39.5-40.0°C                                                    | 7.0  | 9.0  |
| >40.0°C                                                         | 2.5  | 2.0  |

N=total number of participants in the group

\* Post-dose 1 (0-28 days), there was one injection-site rubella-like rash and one injection-site varicella-like rash, and both were reported in the subcutaneous group.

† Intensity of injection site reaction: mild or  $\leq 2.5$  cm; moderate or  $>2.5$  to  $\leq 5.0$  cm; severe or  $>5.0$  cm.

‡ Intensity of pain: mild: awareness of symptom but easily tolerated; moderate: definitely acting like something is wrong; severe: extremely distressed or unable to do usual activities.

§ The percentage of fever is defined within the population who had valid temperature measurements. Three participants in IM group and four participants in SC group did not have temperature measurements and were excluded from the denominator; resulting in N=199 and N=199, respectively.

¶ In the IM Group 96.0% of fevers were documented using the rectal route of measurement and 4.0% of fevers were documented only by the axillary route of measurement. In the SC Group 99.3% of fevers were documented using the rectal route of measurement and 0.7% of fevers were documented only by the axillary route of measurement.

**Table 2: Proportion of Participants Reporting Solicited Adverse Reactions Following Second Vaccination with ProQuad, by the Intramuscular or Subcutaneous Route**

|  | INTRAMUSCULAR | SUBCUTANEOUS |
|--|---------------|--------------|
|--|---------------|--------------|

|                                                           | N=201<br>%   | N=200<br>% |
|-----------------------------------------------------------|--------------|------------|
| Solicited injection-site adverse reactions (Days 0 to 4)* |              |            |
| Erythema <sup>†</sup>                                     | 15.4         | 27.0       |
| Mild                                                      | 13.9         | 22.5       |
| Moderate                                                  | 1.0          | 4.5        |
| Severe                                                    | 0            | 0          |
| Missing                                                   | 0.5          | 0          |
| Pain <sup>‡</sup>                                         | 10.0         | 10.0       |
| Mild                                                      | 9.0          | 7.0        |
| Moderate                                                  | 0.5          | 3.0        |
| Severe                                                    | 0.5          | 0          |
| Swelling <sup>†</sup>                                     | 6.0          | 12.5       |
| Mild                                                      | 5.0          | 11.0       |
| Moderate                                                  | 1.0          | 1.0        |
| Severe                                                    | 0            | 0          |
| Missing                                                   | 0            | 0.5        |
| Solicited systemic adverse reactions (Days 0 to 28)       |              |            |
| Measles-like rash                                         | 0            | 1.0        |
| Rubella-like rash                                         | 2.0% (4/201) | 1.0        |
| Varicella-like rash                                       | 0            | 2.0        |
| Zoster-like rash                                          | 0            | 0          |
| Mumps-like illness                                        | 0.5          | 0          |
| Fever (temperature ≥ 38.0°C) <sup>§,¶</sup>               | 50.0         | 47.2       |
| 38.0-38.5°C                                               | 13.8         | 16.4       |
| >38.5-39.0°C                                              | 18.4         | 10.8       |
| >39.0-39.5°C                                              | 11.2         | 11.3       |
| >39.5-40.0°C                                              | 5.6          | 7.2        |
| >40.0°C                                                   | 1.0          | 1.5        |

N=total number of participants in the group

\* Post-dose 2 (0-28 days), there was one injection-site measles-like rash and this was reported in the subcutaneous group.

† Intensity of injection site reaction: mild or  $\leq 2.5$  cm; moderate or  $>2.5$  to  $\leq 5.0$  cm; severe or  $>5.0$  cm.

‡ Intensity of pain: mild: awareness of symptom but easily tolerated; moderate: definitely acting like something is wrong; severe: extremely distressed or unable to do usual activities.

§ The percentage of fever is defined within the population who had valid temperature measurements. Five participants in IM group and five participants in SC group did not have temperature measurements and were excluded from the denominator; resulting in N=196 and N=195, respectively.

¶ In the IM Group 95.9% of fevers were documented using the rectal route of measurement and 4.1% of fevers were documented only by the axillary route of measurement. In the SC Group 98.9% of fevers were documented using the rectal route of measurement and 1.1% of fevers were documented only by the axillary route of measurement.

Unsolicited adverse events (0-28 days post-vaccination 1 and 2) and serious adverse events (day 0 to last visit) were recorded using diary cards supplemented by medical review. Data on unsolicited adverse events were transcribed into the study database during an on-site visit at day 30 and day 42. Serious adverse events occurred at a rate of 1% and 0.5% in each group post-dose 1 and 2, respectively. None of the serious adverse events that occurred were considered related to the study vaccination.

### *Post-Marketing Observational Safety Surveillance Study*

Safety was evaluated in an observational study that included 69,237 children vaccinated with ProQuad 12 months to 12 years old. A historical comparison group included 69,237 age-, gender-, and date-of-vaccination (day and month)-matched subjects who were given M-M-R II and VARIVAX concomitantly. The primary objective was to assess the incidence of febrile seizures occurring within various time intervals after vaccination in 12- to 60-month-old children who had neither been vaccinated against measles, mumps, rubella, or varicella, nor had a history of the wild-type infections (N=31,298 vaccinated with ProQuad, including 31,043 who were 12 to 23 months old). The incidence of febrile seizures was also assessed in a historical control group of children who had received their first vaccination with M-M-R II and VARIVAX concomitantly (N=31,298, including 31,019 who were 12 to 23 months old). The secondary objective was to assess the general safety of ProQuad in the 30-day period after vaccination in children 12 months to 12 years old.

In pre-licensure clinical studies, an increase in fever was observed 5 to 12 days after vaccination with ProQuad (dose 1) compared to M-M-R II and VARIVAX (dose 1) given concomitantly. In the post-marketing observational surveillance study, results from the primary safety analysis revealed an approximate two-fold increase in the risk of febrile seizures in the same 5 to 12 day timeframe after vaccination with ProQuad (dose 1). The incidence of febrile seizures 5 to 12 days after ProQuad (dose 1) (0.70 per 1000 children) was higher than that in children receiving M-M-R II and VARIVAX concomitantly (0.32 per 1000 children) [relative risk (RR) 2.20, 95% confidence interval (CI): 1.04, 4.65]. The incidence of febrile seizures 0 to 30 days after ProQuad (dose 1) (1.41 per 1000 children) was similar to that observed in children receiving M-M-R II and VARIVAX concomitantly [RR 1.10 (95% CI: 0.72, 1.69)]. See Table 3. General safety analyses revealed that the risks of fever (RR=1.89; 95% CI: 1.67, 2.15) and skin eruption (RR=1.68; 95% CI: 1.07, 2.64) were significantly higher after ProQuad (dose 1) compared with those who received concomitant first doses of M-M-R II and VARIVAX, respectively. All medical events that resulted in hospitalization or emergency room visits were compared between the group given ProQuad and the historical comparison group, and no other safety concerns were identified in this study.

**Table 3**

**Confirmed Febrile Seizures Days 5 to 12 and 0 to 30 After Vaccination with ProQuad (dose 1) Compared to Concomitant Vaccination with M-M-R II and VARIVAX (dose 1) in Children 12 to 60 Months of Age**

| Time period  | ProQuad cohort<br>(N=31,298) |                        | MMR+V cohort<br>(N=31,298) |                        | Relative risk (95% CI) |
|--------------|------------------------------|------------------------|----------------------------|------------------------|------------------------|
|              | n                            | Incidence<br>per 1,000 | n                          | Incidence<br>per 1,000 |                        |
| 5 to 12 days | 22                           | 0.70                   | 10                         | 0.32                   | 2.20 (1.04, 4.65)      |
| 0 to 30 days | 44                           | 1.41                   | 40                         | 1.28                   | 1.10 (0.72, 1.69)      |

In this observational post-marketing study, no case of febrile seizure was observed during the 5 to 12 day post-vaccination time period among 26,455 children who received ProQuad as a second dose of M-M-R II and/or VARIVAX (25,212 as second dose of M-M-R II and VARIVAX, 1,056 as a second dose of M-M-R II, and 187 as a second dose of VARIVAX). In addition, detailed general safety data were available from the 25,212 children who received ProQuad as

a second dose of M-M-R II and VARIVAX, most of them (95%) between 4 and 6 years of age, and an analysis of these data by an independent, external safety monitoring committee did not identify any specific safety concern.

### **XIII. OVERDOSAGE**

Administration of a higher than recommended dose of ProQuad was reported rarely and the adverse reaction profile was comparable to that observed with the recommended dose of ProQuad.

### **XIV. APPEARANCE & AVAILABILITY**

Before reconstitution, ProQuad is a lyophilized plug that appears as white to faint yellow crystalline pellet. ProQuad, when reconstituted, is a pale yellow to light pink liquid.

The vaccine is supplied in glass vials closed by rubber stoppers and capped with aluminium seals and plastic caps.

The vaccine is available as a single-dose vial packed with a vial or syringe of sterile diluent for reconstitution.

Not all pack sizes are marketed.

### **XV. SHELF LIFE**

Please refer to the expiry date on the outer carton.

### **XVI. STORAGE CONDITION**

During shipment, to ensure that there is no loss of potency, the vaccine must be maintained at a temperature of 2° to 8°C (36° to 46°F) or colder, but not exceed temperatures lower than -50°C

(-58°F). Use of dry ice may subject ProQuad to temperatures colder than -50°C (-58°F).

Before reconstitution, store the lyophilized vaccine in a refrigerator at 2° to 8°C (36° to 46°F) or colder. The vaccine may also be stored in a freezer and subsequently transferred to a refrigerator; however, the lyophilized vaccine should not be refrozen.

**DO NOT STORE LYOPHILIZED VACCINE AT ROOM TEMPERATURE.**

**IF LYOPHILIZED VACCINE IS INADVERTENTLY STORED AT ROOM TEMPERATURE, IT SHOULD BE DISCARDED.**

Protect the vaccine from light at all times since such exposure may inactivate the vaccine viruses.

**DISCARD IF RECONSTITUTED VACCINE IS NOT USED WITHIN 30 MINUTES.**

The diluent should be stored separately at room temperature (20° to 25°C, 68° to 77°F), or in the refrigerator (2° to 8°C, 36° to 46°F).

## **XVII. MANUFACTURER**

### **MERCK SHARP & DOHME LLC**

Sumneytown Pike,  
West Point, PA 19486, U.S.A.

## **XVIII. PACKER AND RELEASER**

### **Merck Sharp & Dohme BV**

Waarderweg 39,  
2031 BN, Haarlem,  
Netherlands.

## **XIX. PRODUCT REGISTRATION HOLDER**

**MERCK SHARP & DOHME (MALAYSIA) SDN. BHD.**

Lot No. B-22-1 & B-22-2, Level 22

The Ascent, Paradigm No.1

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